

Package ‘PhyloTraitDynamics’

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Title Companion Code for Phylogenetic Trait Dynamics

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Description Conservative companion package for reproducing numerical results and figures for an article on birth-death processes, MRCA ages, and Brownian empirical moments on phylogenies. The package exposes a small public API with package-local implementations.

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birth_death_brownian_simulate

Simulate Birth-Death Brownian Empirical Moments

Description

Simulates birth-death Brownian paths and records, on the requested time grid, the number of living lineages, the empirical mean of their traits, and their empirical variance.

Usage

```
birth_death_brownian_simulate(
  birth,
  death,
  sigma2 = 1,
  time_end,
  time_step,
  B,
  x0 = 0,
  seed = NULL,
  n_envelope = 4000,
  safety_factor = 1.05
)
```

Arguments

birth	Non-negative birth rate, either a numeric constant or a function of time.
death	Non-negative death rate, either a numeric constant or a function of time.
sigma2	Brownian variance parameter.
time_end, time_step	Time grid definition.
B	Number of simulated paths.
x0	Initial trait value.
seed	Optional random seed.
n_envelope	Number of points used by the thinning envelope.
safety_factor	Multiplicative safety factor for the thinning envelope.

Value

An object of class "birth_death_brownian_simulation" containing the time grid, matrices of empirical means, empirical variances and lineage counts, and a simulation summary.

empirical_mean_compute_variance

Variance of the Empirical Mean Under Birth-Death Brownian Dynamics

Description

Computes the theoretical variance of the empirical mean of Brownian traits, conditional on survival at each time.

Usage

```
empirical_mean_compute_variance(
  birth,
  death,
  sigma2 = 1,
  time_end,
  time_step,
  n_steps = 1200,
  tol_sing = 1e-07,
  li2_tol = 1e-12
)
```

Arguments

birth	Non-negative birth rate, either a numeric constant or a function of time.
death	Non-negative death rate, either a numeric constant or a function of time.
sigma2	Brownian variance parameter.
time_end, time_step	Time grid definition.
n_steps	Number of integration steps for numerical theory.
tol_sing, li2_tol	Numerical tolerances passed to the original implementation.

Value

A data frame with time, survival probability, empirical mean variance, and conditioning columns. The original script result is attached as attribute "original_result".

empirical_mean_plot_paths

Plot Empirical Mean Paths

Description

Plots individual simulated empirical mean paths.

Usage

```
empirical_mean_plot_paths(
  simulation,
  max_paths = 100,
  alpha = 0.08,
  main = NULL,
  xlab = "time",
  ylab = "Empirical mean"
)
```

Arguments

simulation An "empirical_mean_simulation" object.
 max_paths Maximum number of simulated paths to draw.
 alpha Transparency used for simulated paths.
 main, xlab, ylab Base graphics labels.

Value

Invisibly returns simulation.

empirical_mean_plot_variance

Plot Variance of the Empirical Mean

Description

Plots Monte Carlo summaries and/or theoretical values for the variance of the empirical mean.

Usage

```
empirical_mean_plot_variance(
  simulation = NULL,
  theory = NULL,
  summary = NULL,
  main = NULL,
  xlab = "time",
  ylab = "Variance of empirical mean"
)
```

Arguments

simulation Optional "empirical_mean_simulation" object.
 theory Optional data frame returned by empirical_mean_compute_variance().
 summary Optional data frame returned by empirical_mean_summarise_simulation().
 If omitted and simulation is supplied, it is computed automatically.
 main, xlab, ylab Base graphics labels.

Value

Invisibly returns a list containing simulation, theory, and summary.

empirical_variance_compute_expectation

Expected Empirical Variance Under Birth-Death Brownian Dynamics

Description

Computes the theoretical expectation of the empirical trait variance on a time grid for a birth-death process with Brownian trait evolution.

Usage

```
empirical_variance_compute_expectation(
  birth,
  death,
  sigma2 = 1,
  time_end,
  time_step,
  conditioning = c("none", "survival"),
  method = c("auto", "numeric", "closed_constant"),
  n_steps = 2000,
  tol = 1e-08,
  constant_tol = 1e-10,
  li2_tol = 1e-12
)
```

Arguments

birth	Non-negative birth rate, either a numeric constant or a function of time.
death	Non-negative death rate, either a numeric constant or a function of time.
sigma2	Brownian variance parameter.
time_end, time_step	Time grid definition.
conditioning	Conditioning used for the expectation. "none" is unconditional and "survival" conditions on $N(t) > 0$.
method	Theory evaluation method. "auto" uses the closed constant-rate formula when the birth and death rates are constant on the time grid, and otherwise uses the numerical method. "closed_constant" requires constant birth and death rates.
n_steps	Number of integration steps for numerical theory.
tol, constant_tol, li2_tol	Numerical tolerances passed to the original implementation.

Value

A data frame with time, survival probability, expected empirical variance, conditioning, and method columns. The original script result is attached as attribute "original_result".

empirical_variance_plot_expectation

Plot Expected Empirical Variance

Description

Plots simulated empirical variance paths, Monte Carlo summaries, and/or the theoretical expectation on a common set of axes.

Usage

```
empirical_variance_plot_expectation(
  simulation = NULL,
  theory = NULL,
  summary = NULL,
  show_paths = TRUE,
  conditioning = c("none", "survival"),
  max_paths = Inf,
  alpha = 0.05,
  main = NULL,
  xlab = "time",
  ylab = "Empirical variance"
)
```

Arguments

simulation	Optional "empirical_variance_simulation" object.
theory	Optional data frame returned by empirical_variance_compute_expectation().
summary	Optional data frame returned by empirical_variance_summarise_simulation(). If omitted and simulation is supplied, it is computed automatically.
show_paths	Logical; whether to draw individual simulated paths.
conditioning	Which expectation to plot.
max_paths	Maximum number of simulated paths to draw.
alpha	Transparency used for simulated paths.
main, xlab, ylab	Base graphics labels.

Value

Invisibly returns a list containing simulation, theory, and summary.

```
empirical_variance_plot_paths
```

Plot Empirical Variance Paths

Description

Plots individual simulated empirical variance paths.

Usage

```
empirical_variance_plot_paths(  
  simulation,  
  max_paths = 100,  
  alpha = 0.08,  
  main = NULL,  
  xlab = "time",  
  ylab = "Empirical variance"  
)
```

Arguments

<code>simulation</code>	An "empirical_variance_simulation" object.
<code>max_paths</code>	Maximum number of simulated paths to draw.
<code>alpha</code>	Transparency used for simulated paths.
<code>main, xlab, ylab</code>	Base graphics labels.

Value

Invisibly returns `simulation`.

```
fixed_tree_compute_theoretical_summary
```

Theoretical Summary on a Fixed Tree

Description

Computes theoretical summaries of empirical mean variance and empirical variance for Brownian traits on a fixed phylogenetic tree.

Usage

```
fixed_tree_compute_theoretical_summary(  
  tree,  
  sigma2 = 1,  
  time_start = 0,  
  time_end = NULL,  
  time_step  
)
```

Arguments

tree	A phylogenetic tree of class "phylo".
sigma2	Brownian variance parameter.
time_start	Start of the time grid.
time_end	End of the time grid. If NULL, the maximum tree time is used.
time_step	Time step used to define the time grid.

Value

A data frame with time, lineage counts, empirical mean variance, empirical variance expectation, empirical variance quartiles, and empirical variance variance. The tree, sigma2, and original result are attached as attributes.

fixed_tree_plot_brownian_realization

Plot a Fixed-Tree Brownian Realization

Description

Plots the simulated Brownian realization together with the fixed tree using the original plotting routine.

Usage

```
fixed_tree_plot_brownian_realization(
  x,
  n_bands = 10,
  time_band_col = grDevices::adjustcolor("blue", alpha.f = 0.05)
)
```

Arguments

x	A fixed_tree_brownian_realization object.
n_bands	Number of time bands shown by the original plotting routine.
time_band_col	Color used for time bands.

Value

Invisibly returns x.

fixed_tree_plot_theoretical_summary

Plot a Fixed-Tree Theoretical Summary

Description

Plots fixed-tree theoretical summaries for empirical mean variance and empirical variance, optionally with the tree.

Usage

```
fixed_tree_plot_theoretical_summary(
  x,
  tree = NULL,
  show_tree = TRUE,
  main = NULL,
  time_band_col = grDevices::adjustcolor("blue", alpha.f = 0.05),
  band_col = grDevices::adjustcolor("red", alpha.f = 0.18),
  expectation_col = "blue",
  median_col = "red",
  mean_variance_col = "darkgreen"
)
```

Arguments

<code>x</code>	A <code>fixed_tree_theoretical_summary</code> object, as returned by <code>fixed_tree_compute_theoretical_summary</code> .
<code>tree</code>	Optional phylogenetic tree of class "phylo". If NULL, the tree attached to <code>x</code> is used when available.
<code>show_tree</code>	Logical; whether to show the tree panel.
<code>main</code>	Optional main title.
<code>time_band_col</code>	Color used for time bands.
<code>band_col</code>	Color used for the empirical-variance interquartile band.
<code>expectation_col</code>	Color used for the empirical-variance expectation.
<code>median_col</code>	Color used for the empirical-variance median and quartiles.
<code>mean_variance_col</code>	Color used for the empirical-mean variance.

Value

Invisibly returns `x`.

fixed_tree_simulate_brownian_realization

Simulate Brownian Traits on a Fixed Tree

Description

Simulates one Brownian trait realization on a fixed phylogenetic tree and records empirical mean and empirical variance through time.

Usage

```
fixed_tree_simulate_brownian_realization(
  tree,
  sigma2 = 1,
  time_step,
  seed = NULL
)
```

Arguments

tree	A phylogenetic tree of class "phylo".
sigma2	Brownian variance parameter.
time_step	Time step used for the simulation output.
seed	Optional random seed.

Value

An object of class "fixed_tree_brownian_realization" containing the input tree, simulation data frame, time series, parameters, and seed.

mrca_age_compute_distribution

MRCA Age Distribution at One Time

Description

Computes the distribution and summary statistics of the most recent common ancestor age at a fixed time under a birth-death process.

Usage

```
mrca_age_compute_distribution(
  birth,
  death,
  time,
  time_step = NULL,
  probs = c(0.25, 0.5, 0.75),
  n_internal = NULL
)
```

Arguments

birth	Non-negative birth rate, either a numeric constant or a function of time.
death	Non-negative death rate, either a numeric constant or a function of time.
time	Observation time.
time_step	Internal time step. If NULL, a default step is derived from time.
probs	Quantile probabilities to report.
n_internal	Optional number of internal subintervals; when supplied it overrides time_step.

Value

An object of class "mrca_age_distribution" with distribution grid, summary statistics, requested probabilities, and the original result.

mrca_age_compute_dynamics

MRCA Age Distribution Dynamics

Description

Computes MRCA age distribution summaries over a time grid under a birth-death process.

Usage

```
mrca_age_compute_dynamics(
  birth,
  death,
  time_end,
  time_step,
  probs = c(0.25, 0.5, 0.75),
  n_internal_per_step = 10L,
  include_zero = TRUE
)
```

Arguments

birth	Non-negative birth rate, either a numeric constant or a function of time.
death	Non-negative death rate, either a numeric constant or a function of time.
time_end, time_step	Time grid definition.
probs	Quantile probabilities to report.
n_internal_per_step	Number of internal integration substeps per output time step.
include_zero	Logical; whether to include time zero in the original dynamics computation.

Value

An object of class "mrca_age_dynamic" containing the summary data frame, distribution grids, probabilities, and original call.

`mrca_age_plot_dynamics`*Plot MRCA Age Distribution Dynamics*

Description

Plots MRCA age summaries over time, including median, optional interquartile band, and optional expectation.

Usage

```
mrca_age_plot_dynamics(  
  x,  
  show_iqr = TRUE,  
  show_expectation = TRUE,  
  main = NA,  
  xlab = "Time",  
  ylab = "MRCA age",  
  add_legend = TRUE,  
  ...  
)
```

Arguments

<code>x</code>	An "mrca_age_dynamic" object.
<code>show_iqr</code>	Logical; whether to draw the interquartile band.
<code>show_expectation</code>	Logical; whether to draw the expectation curve.
<code>main</code> , <code>xlab</code> , <code>ylab</code>	Base graphics labels.
<code>add_legend</code>	Logical; whether to draw the plot legend.
<code>...</code>	Additional arguments passed to <code>graphics::plot()</code> .

Value

Invisibly returns `x`.

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